

Section - I: Multiple Choice Questions

1. What is a common indication of overfitting in a machine learning model?
 - A) The model performs well on both the training and testing datasets.
 - B) The model performs well on the training dataset but poorly on the testing dataset.
 - C) The model performs poorly on both the training and testing datasets.
 - D) The model requires a large amount of data to make predictions.

Section - II: Descriptive / Case Study-Based Questions

1. A marketing firm, "Smart Ads," wants to develop a machine learning model to predict the click-through rate (CTR) of their online advertisements. They collect a large dataset of features, including ad content, user demographics, and browsing history. After training a complex neural network model on the dataset, the model achieves a high accuracy of 95% on the training set. However, when tested on a holdout validation set, the accuracy drops to 60%. What might be the reason for this significant decrease in accuracy, and how would you modify the model to improve its performance on the validation set?